STARTUP FAILURE ANALYSIS

A Data-Driven Investigation into Why Startups Fail

Data Source: Failory.com
Dataset: 186 failed startups across 18 countries
Methods: Web scraping, EDA, KNN classification, Linear regression
Tools: Python, BeautifulSoup, pandas, scikit-learn, matplotlib

1. Introduction

Startup failure is one of the most studied yet least fully understood phenomena in entrepreneurship. Statistics consistently show that more than 90% of startups fail within their first few years, yet each failure has a distinct story shaped by the industry, funding environment, team composition, and market conditions of its time.

This project investigates startup failure patterns using data collected from Failory.com, a platform that publishes post-mortem analyses and founder interviews for failed startups. The goal is to move beyond anecdotal failure stories and identify systematic, data-driven patterns in how and why startups shut down.

The project covers three main areas: (1) data collection through web scraping, (2) exploratory data analysis and visualisation, and (3) machine learning models to predict failure causes and startup lifespan. The findings are intended to serve both as an academic analysis and as a practical reference for founders and investors.

1.1 Research Questions

This project attempts to answer the following questions:

- What are the most common causes of startup failure, and how do they differ across industries?
- Does the amount of funding raised influence how long a startup survives?
- Can a machine learning model predict the likely cause of failure given basic startup attributes?
- What does a startup's lifespan distribution look like, and what factors correlate with longer survival?

2. Data Collection

2.1 Data Source

All data was collected from Failory.com, which maintains one of the largest publicly available databases of failed startups. The site organises startups by country, industry, business model, and cause of failure, making it well-suited for structured web scraping.

Failory publishes individual case-study pages for each failed startup. Each page contains structured fields including the startup name, founders, country of origin, industry, year founded, year closed, team size range, funding amount range, and the stated primary cause of failure as reported by the founders themselves.

2.2 Scraping Process

The scraping pipeline was built in Python using the requests and BeautifulSoup libraries. The process followed these steps:

- Scraped the main Failory failures directory page to extract all article links grouped by category (country, industry, cause).

- Selected 19 specific category pages covering 8 countries and 11 industries to ensure diverse geographic and sectoral coverage.
- For each category page, extracted all startup blocks.
- Parsed each startup block by splitting on the separator and extracting fields by matching keyword patterns (e.g., "country:", "industry:", "funding amount:").

2.3 Dataset Overview

The final dataset contains 186 startup records collected across categories spanning countries including Germany, India, Belgium, Russia, Bulgaria, Spain, Sweden, and Canada, and industries including Fashion, Education, AI, E-Commerce, Crypto, FoodTech, and more.

Metric	Value
Total records	186 startups
Columns	10 (Startup, Founders, Country, Industry, Started In, Closed In, Employees, Funding Amount, Failure Cause, Description)
Year range	1994 to 2024
Countries covered	18
Industries covered	15+
Missing values	Employees: 95 missing (51%) | All other columns: complete

3. Data Cleaning and Preprocessing

3.1 Lifetime Feature Engineering

The dataset did not contain a direct "lifespan" column. This was computed by subtracting the "Started In" year from the "Closed In" year. A small number of rows had "Active" in the Closed In field — these were replaced with 2026 and treated as right-censored observations (still running at time of data collection).

Both year columns were converted to numeric type after handling these edge cases. The resulting Lifetime column represents the number of full years the startup operated before shutting down.

3.2 Funding Amount Cleaning

Funding amounts were provided as ranges or inequality expressions rather than exact values (e.g., "$1M-$10M", "> $50M", "< $100K"). This is a common issue with startup datasets, as precise funding figures are often not publicly disclosed.

The cleaning function applied the following rules:

- Range values (e.g., "$1M-$10M"): converted both endpoints to a common unit (millions USD) and took the arithmetic mean. For example, "$1M-$10M" became 5.5.
- Inequality values (e.g., "> $50M", "< $100K"): the bound value was extracted and used directly as a point estimate.
- Zero or unknown values ("$0", "No Data"): treated as 0 or NaN respectively.
- Remaining NaN values were imputed with the column median to preserve all rows for modelling.

3.3 Failure Cause Grouping

The raw dataset contained 18 distinct failure cause labels as reported by founders. Many of these were semantically related but labelled differently, resulting in class imbalance that would hinder machine learning. A mapping was applied to consolidate them into 6 broader groups:

Group	Original Labels Included	Count
Market	No Market Need, Bad Market Fit, Competition, Bad Marketing, Lack of PMF	69 (37%)
Management	Bad Management, Lack of Experience, Lack of Focus, Bad Business Model, Failure to Pivot	52 (28%)
Financial	Lack of Funds, Mismanagement of Funds	21 (11%)
External	Legal Challenges, Dependence on Others, Acquisition Flu	21 (11%)
Product	Poor Product	10 (5%)
Other	Multiple Reasons, Other	13 (7%)

Market-related failures are the most common, representing 37.1% of all cases, followed by Management failures at 28.0%. Together these two groups account for nearly two-thirds of all startup failures in the dataset.

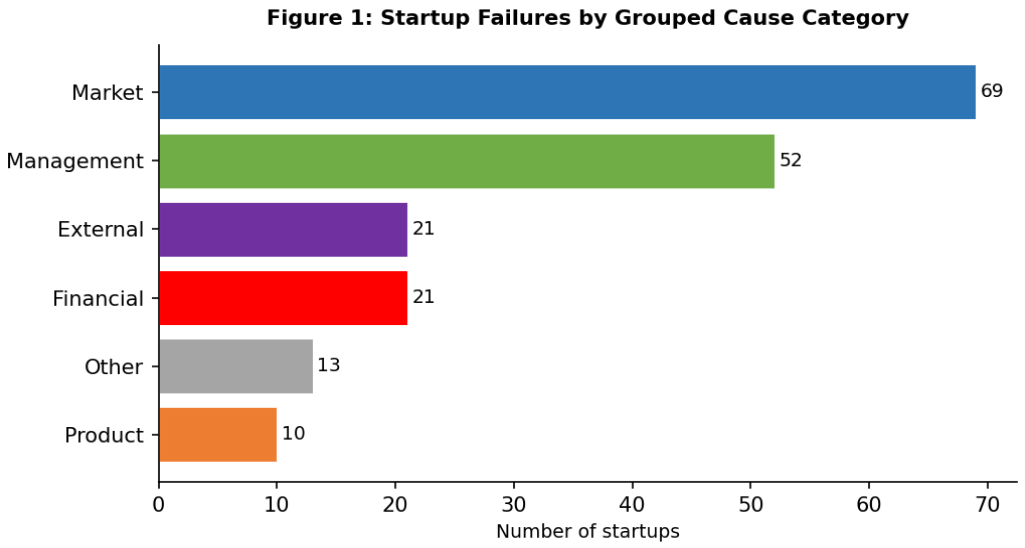
3.4 Employee Field

The Employees column had 95 missing values (51% of records), making it unsuitable as a reliable regression feature. It was retained for descriptive analysis and visualisation but excluded from the final predictive models to avoid introducing imputation bias across half the dataset.

4. Exploratory Data Analysis

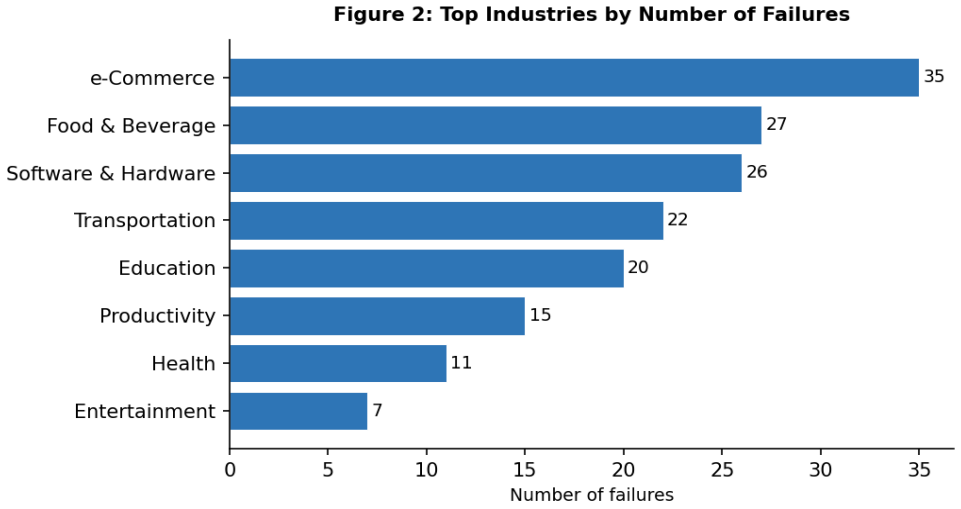
4.1 Failure Cause Distribution

The chart below shows the count of startups in each grouped failure cause category. Market failures dominate, followed closely by Management failures. Financial failures, while commonly cited in popular discourse, rank third and account for only 11% of cases — suggesting that most startups that failed did not simply run out of money, but rather faced more fundamental market or execution problems.



4.2 Failures by Industry

E-Commerce leads all industries with 35 failures, followed by Food & Beverage (27), Software & Hardware (26), and Transportation (22). It is important to note that these counts reflect the volume of failures in Failory's dataset rather than the failure rate within each industry — sectors with more startup activity will naturally produce more absolute failures.

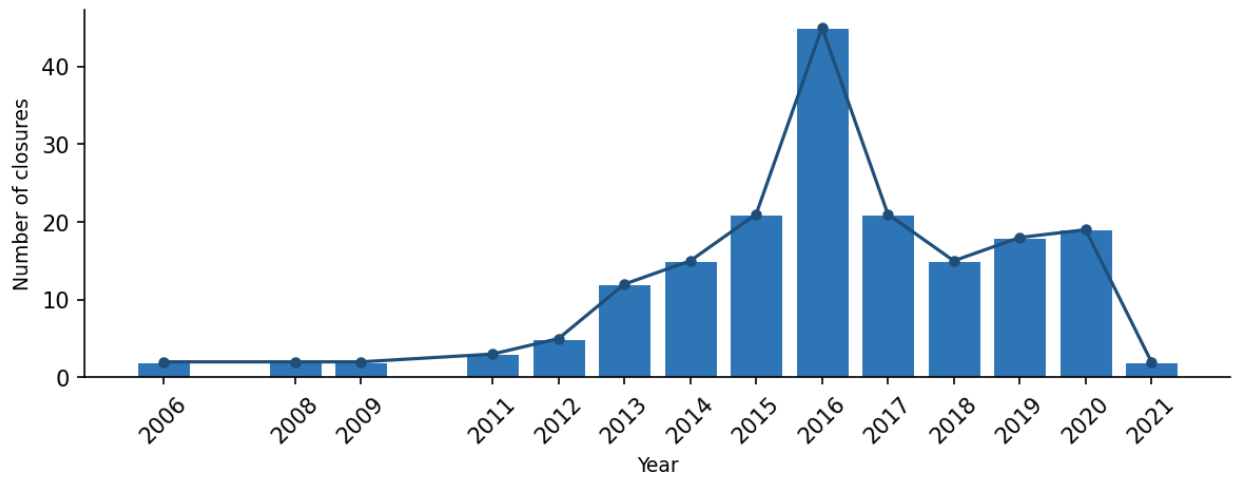


4.3 Failures Over Time

The temporal distribution of closures shows a notable clustering in the 2015-2020 period. This reflects both the surge in startup activity following the 2010-2013 funding boom and the typical 3-5 year runway before a startup exhausts its resources. The spike around 2016-2018 is consistent with the cohort of startups founded during the 2012-2015 accelerator and seed-funding expansion period.

Recent years (2022-2024) appear underrepresented because Failory's documentation of recent failures lags the actual closure events.

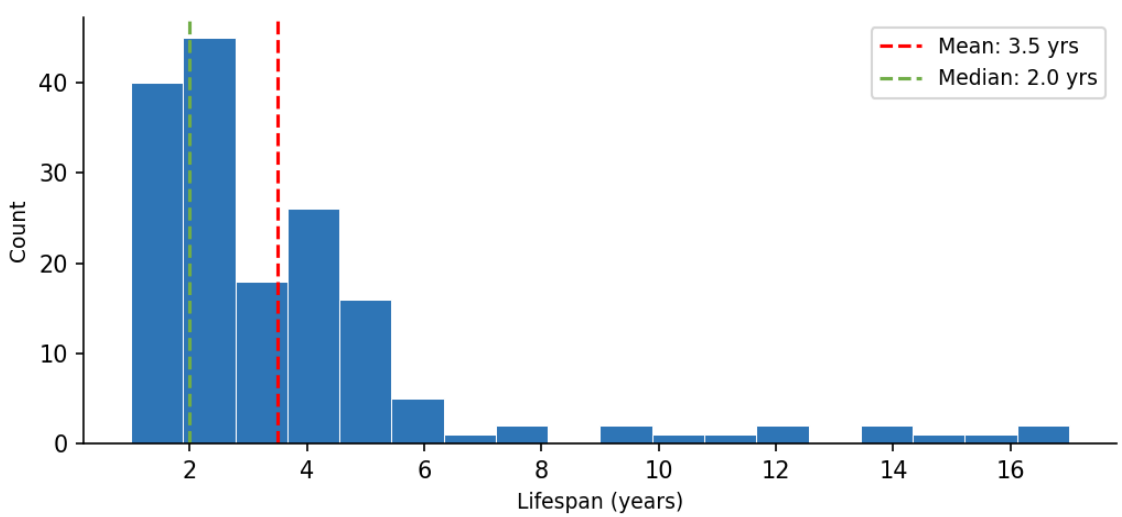
Figure 3: Startup Failures Over Time (2005–2024)



4.4 Startup Lifespan Distribution

The lifespan histogram reveals a strongly right-skewed distribution. The median lifespan is 2 years, meaning half of all failed startups in the dataset closed within their first two years of operation. The mean is 3.5 years, pulled upward by a small number of long-running outliers (startups that operated for 10 or more years before shutting down).

Figure 4: Distribution of Startup Lifespan (Years)

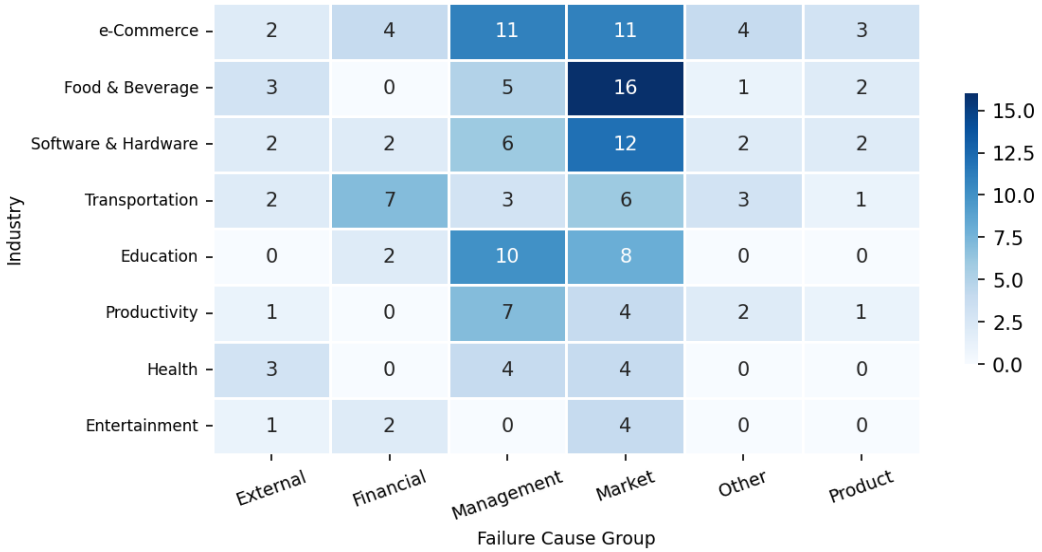


4.5 Industry versus Failure Cause Heatmap

The heatmap below cross-tabulates the top 8 industries against all 6 grouped failure causes. Darker cells indicate more startups in that industry-cause combination. Several industry-specific patterns emerge:

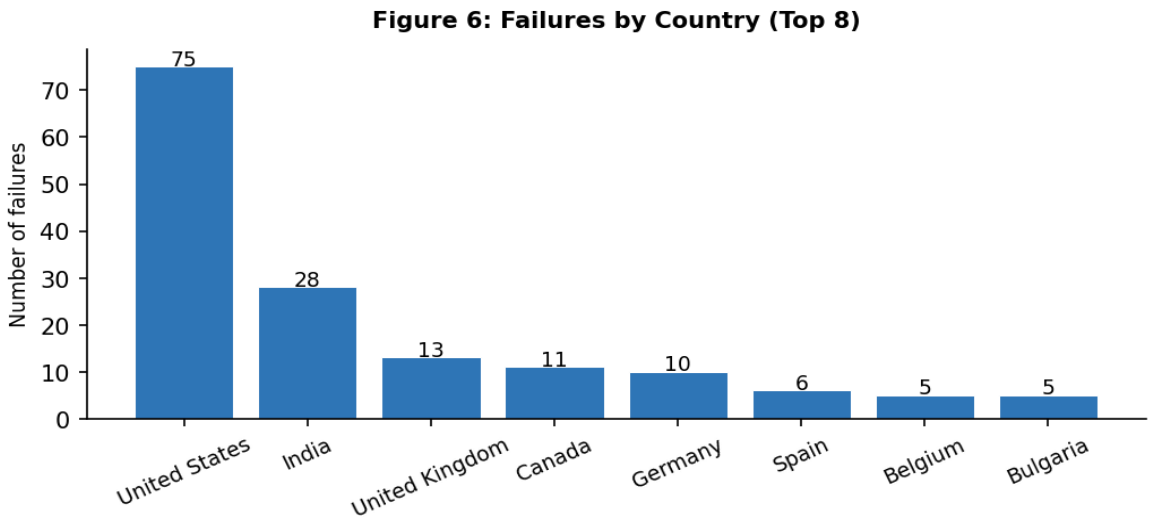
- E-Commerce failures are spread across Market and Management causes, with Competition being a major driver.
- Transportation failures are disproportionately concentrated in the Financial and External categories, consistent with the capital-intensive nature of the sector.
- Education failures cluster heavily in Market causes, reflecting the difficulty of finding a sustainable business model in EdTech.
- Software & Hardware failures are more evenly distributed, suggesting diverse failure pathways.

Figure 5: Industry vs. Failure Cause Heatmap (Top 8 Industries)



4.6 Geographic Distribution

The country distribution shows a strong skew towards the United States. This suggests that the dataset is not globally balanced and primarily reflects US startup ecosystems. Therefore, conclusions drawn from this dataset may not generalize well to other countries.



5. Machine Learning Models

5.1 KNN Classifier — Predicting Failure Cause

5.1.1 Problem Setup

A K-Nearest Neighbours (KNN) classifier was trained to predict the grouped failure cause (Market, Management, Financial, External, Product, Other) given three input features: funding amount (numeric midpoint), industry (one-hot encoded), and startup lifespan in years.

The train-test split was 80/20 with stratification on the target class to ensure proportional representation of each cause category in both sets. StandardScaler was applied to the numeric features (funding and lifespan) before distance computation, as KNN is sensitive to the scale of its input features.

5.1.2 Feature Encoding

Proper encoding of the categorical features was critical for KNN performance. The key decisions were:

- Industry: One-hot encoded using pandas get_dummies(). This prevents KNN from treating industries as ordinal numbers and computing meaningless distances between them.
- Funding: Converted to numeric midpoint, then scaled with StandardScaler so that a $1M difference in funding has comparable weight to a 1-year difference in lifespan after scaling.
- Lifespan: Scaled with StandardScaler alongside funding to ensure neither feature dominates the distance computation.

Class	Precision	Recall	F1-Score	Support
Financial	0.29	0.50	0.36	4
Market	0.50	0.73	0.59	11
Management	0.50	0.25	0.33	12
External	0.50	0.50	0.50	4
Product	0.33	0.25	0.29	4
Other	0.00	0.00	0.00	3
Weighted avg	0.42	0.42	0.40	38

The overall test accuracy was 42%, achieved with k=8. This is 2.5 times better than the random baseline of 16.7% (1/6 for six classes), indicating that the model has captured real predictive signal in the features.

The "Market" class had the strongest recall (0.73), likely because it is the largest and most represented class. "Other" received zero predictions — these classes have only 3-4 test samples each, which is insufficient for KNN to learn reliable boundaries.

5.1.3 Limitations and Improvements

The modest accuracy reflects three structural challenges:

- Small dataset (186 rows): KNN requires enough samples in each class to form meaningful cluster boundaries. With 6 classes and some having fewer than 10 samples, the model cannot generalise well.
- Class imbalance: Market (37%) and Management (28%) together dominate the dataset.
- Feature limitations: Funding, industry, and lifespan are weakly predictive of failure cause because the same type of startup can fail for many different reasons. Adding features like founding team size, business model type, or year of founding would improve prediction.
- KNN and categorical features: Despite one-hot encoding, KNN still struggles with high-dimensional sparse feature spaces produced by many industry dummies.

5.2 Linear Regression — Predicting Startup Lifespan

5.2.1 Problem Setup

A linear regression model was trained to predict startup lifespan (in years) using funding amount and industry as features. The target variable was log-transformed using log1p(Lifetime) before fitting, as the raw lifespan distribution was strongly right-skewed and violated the normality assumption of linear regression residuals.

An 80/20 train-test split was used. Industry was one-hot encoded. Funding was used as the numeric midpoint.

5.2.2 Why Log-Transform the Target?

The raw lifespan distribution is right-skewed, with most startups failing within 1-3 years and a small number surviving for 10-17 years. Fitting linear regression directly on skewed targets produces heteroscedastic residuals — the model's errors grow larger for longer-lived startups.

Applying log1p(Lifetime) converts the distribution into an approximately symmetric bell curve, which satisfies the assumption of normally distributed residuals and makes the coefficient estimates more stable and interpretable.

5.2.3 Results

Metric	Value	Interpretation
R^2 Score	0.396	39.6% of variance in log(lifespan) explained by funding and industry
MAE	0.386 years	Average prediction error in actual years (back-transformed)

The R^2 of 0.396 should be interpreted carefully. With only two features — funding and industry — explaining 39.6% of the variance in how long a startup survives is a meaningful result. The remaining 60.4% is driven by factors this dataset cannot capture: founder quality, market timing, competitive dynamics, and execution ability.

The residual plot confirmed that after log-transformation, residuals are approximately randomly distributed around zero, though some heteroscedasticity remains at the extremes. One outlier with a residual near 10 represents a startup that survived far longer than the model predicted given its ordinary funding and industry profile.

5.2.4 Coefficient Interpretation

The linear regression coefficients indicate which industry features are associated with longer or shorter predicted lifespans, holding funding constant:

- Positive coefficients (longer survival): Transportation, Health, and FinTech industries are associated with longer lifespans, consistent with the higher capital requirements and regulatory barriers that slow but extend these companies.
- Negative coefficients (shorter survival): Social Media, E-Commerce, and Media show negative coefficients, consistent with the fast-paced, winner-take-all dynamics in consumer internet sectors.
- Funding coefficient: Funding has a small positive coefficient, confirming that more capital is associated with longer survival, but the effect size is modest — the type of industry matters more than the amount raised.

6. Key Findings

6.1 Summary of Results

Metric	Value
Most common failure cause	Market failures (37%) — No Market Need, Competition, Bad Market Fit
Second most common	Management failures (28%) — Bad Business Model, Lack of Experience
Median startup lifespan	2.0 years (mean: 3.5 years)
Peak failure years	2015–2020 (lagging the 2012–2015 funding boom)
Top failure industry	E-Commerce (35 failures)
Top country	United States (75 records, 40% of dataset)
KNN accuracy	42.1% (vs. 16.7% random baseline — 2.5x improvement)
Regression R^2	0.396 using funding and industry as predictors
Funding vs lifespan	Weak positive correlation — more funding helps but does not guarantee survival

6.2 Finding 1 — Most Failures are Avoidable

The dominance of Market and Management failures (65% combined) over Financial failures (11%) is one of the most important findings of this analysis. It challenges the common narrative that startups fail because they "run out of money." The data shows that the more fundamental causes are strategic — building a product no one needs, failing to find product-market fit, or having a business model that does not work.

6.3 Finding 2 — Industry Shapes the Type of Failure

The heatmap analysis revealed that failure cause is not uniformly distributed across industries. Transportation startups disproportionately fail due to Financial and External causes, reflecting their capital intensity and regulatory exposure. E-Commerce startups fail primarily due to Competition and Market causes, reflecting the low barriers to entry and high competitive density of that sector.

This has practical implications: an investor evaluating a Transportation startup should focus on the funding runway and regulatory risk, while the same investor looking at an E-Commerce startup should focus primarily on differentiation and customer acquisition costs.

6.4 Finding 3 — The 2-Year Danger Zone

The median lifespan of 2 years identifies what can be called the "2-year danger zone" for startups. Half of all failed startups in this dataset closed within their first two years, suggesting that the critical period for a startup's survival is the early growth phase — roughly months 12-24 — when the initial product has launched but the company has not yet achieved sustainable revenue.

Startups that survive past 5 years show a very different profile: they typically have more funding, operate in capital-intensive industries, or have found a niche market where competition is limited.

6.5 Finding 4 — Funding is Necessary but Not Sufficient

The scatter plot of funding vs. lifespan shows a wide spread with no clear linear trend, confirmed by the weak coefficient in the regression model. While there are no long-surviving startups in the dataset with zero funding, there are also numerous heavily funded startups that failed within 1-2 years.

This suggests that funding extends the runway but does not solve the underlying problem. A startup with a bad market fit will simply fail more expensively if it raises more money.

What the Models Tell Us

The KNN classifier, despite its modest 42.1% accuracy, confirms a fundamental truth: failure cause is hard to predict from external attributes alone. Funding, industry, and lifespan explain some of the variance, but the specific way a startup fails is shaped largely by decisions and circumstances that no public dataset can fully capture.

The regression model's R^2 of 0.396 tells a similar story for lifespan prediction. Funding and industry explain about one-third of why some startups survive longer than others. The remaining two-thirds is the "human factor" — the quality of the team, the timing of their bets, and the adaptability of their strategy in the face of adversity.

These modest model performances are not failures of methodology — they are an honest quantification of how much external, observable data can tell us about the internal dynamics of startup success and failure.

8. Conclusion

This project conducted a complete data analysis pipeline on 186 failed startups scraped from Failory.com — from raw web scraping through data cleaning, exploratory analysis, visualisation, and machine learning modelling.

The key conclusion is straightforward but important: most startups fail not because they run out of money, but because they build something the market does not need or because they cannot execute on a viable business model. Market and management failures together account for 65% of all cases in this dataset, significantly outweighing financial failures at 11%.

The typical failed startup operates for 2 years before closing, regardless of industry or funding level. Funding extends the runway but does not address the underlying strategic problems that cause failure.

The machine learning models produced results consistent with the limited predictive power of external startup attributes. The KNN classifier achieved 42.1% accuracy (2.5x random), and the linear regression explained 39.6% of variance in startup lifespan. Both results represent genuine signal rather than noise, while also confirming that the internal dynamics of startup failure — team quality, strategic adaptability, market timing — cannot be fully captured by a public dataset.

This analysis provides a quantitative foundation for thinking about startup failure and offers a reusable data pipeline for anyone wishing to extend it with additional data sources or more sophisticated modelling approaches.